

[DC] IMMERCOCG: Impact of Extreme Environments on Remote Operations in Critical Systems

Elena López-Contreras*

Fédération ENAC ISAE-SUPAERO ONERA

Université de Toulouse, France

ABSTRACT

Remote operation technologies play a critical role in fields such as medicine, defense, and space exploration, yet existing research often relies on controlled laboratory simulations that fail to capture the complexities of real-world environments. This doctoral project, IMMERCOCG, investigates the impact of extreme conditions—such as underwater environments and real flight—on the psychophysiological states and performance of operators controlling remote systems. This research bridges the gap between controlled studies and real-world applications, aiming to advance human-machine interaction (HMI) for remote operations under stress.

The project involves three experimental campaigns conducted using the same interface: a laboratory study to establish baseline operator performance using motion simulation, underwater teleoperation at varying depths to evaluate stress and cognitive impact, and in-flight teleoperation to address the challenges of controlling robots from a moving aircraft. Preliminary results highlight significant learning effects across tasks and notable performance differences between manual and autonomous control modes. These findings provide actionable insights for designing adaptive HMIs tailored to extreme conditions, contributing to the development of robust and efficient teleoperation systems.

Index Terms: Teleoperation, Extreme Environments, Remote Operations, Interface, Human-Machine Interaction

1 INTRODUCTION

The advancement of remote operation technology has revolutionized critical domains such as medicine, defense, industrial automation, and space exploration. In extreme environments, such as space, deep-sea exploration, or disaster zones, teleoperation enables tasks to be performed remotely, mitigating risks to human operators [3]. Despite significant progress, much of the existing research on teleoperation is grounded in controlled laboratory simulations, which do not fully replicate the stressors and operational challenges present in real-world environments [1, 2].

In reality, teleoperation is seldom performed in calm, predictable conditions—operators often face extreme environments like underwater depths or dynamic flight settings, where physiological stress and operational demands can significantly impact performance. This gap in understanding limits the design and performance optimization of human-robot systems operating in extreme conditions.

IMMERCOCG spans three distinct experimental domains to capture a comprehensive view of teleoperation under varying conditions. First, the study establishes a baseline through controlled laboratory simulations where operators teleoperate a rover using a web interface and motion platforms in a low-stress, controlled environment. This provides a reference point for comparing performance and stress levels under more challenging conditions.

*e-mail: elena.lopez-contreras-gonzalez@isae-supero.fr

Importantly, the perception and control tasks remain consistent across all three environments, with the same web interface and rover used in each scenario—whether the robot is operating automatically or manually controlled by the operator.

The second domain explores underwater teleoperation, where operators control a rover at varying depths (2 m and 15 m) to assess the impact of pressure, immersion, and the potential cognitive load associated with these conditions.

Finally, the research investigates flight-based teleoperation, where operators control the same robot, while onboard a moving aircraft. This scenario introduces additional complexities, including the effects of aircraft motion and varying load factors.

2 METHODS

The core of the IMMERCOCG study is the interface used by participants to control the robot across various experimental conditions.

In the IMMERCOCG case, the interface consists of a web-based platform paired with a gamepad for manual control, as can be seen in Figure 1. For both the manual control and automatic control tasks, the same web interface, task, and robot are used, ensuring consistency across all environments.

The robot's control is divided into two modes: manual and automatic. In manual control mode, participants use the gamepad's A/B/X/Y buttons to control the robot's movements, which can only move forward, backward, and rotate. Each button press moves the robot in one-second increments, providing precise but limited control. Participants also receive visual feedback via a live camera and a live map on the interface.

Additionally, predefined touchpoints (P1-P6) are established to standardize task initiation. They ensure that each participant begins the task from a consistent position.

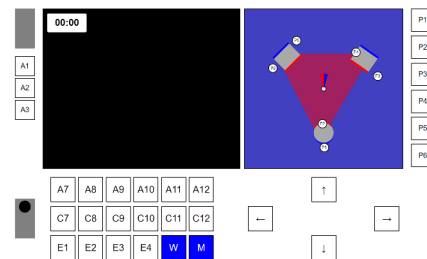


Figure 1: IMMERCOCG Web-based interface

The experiment has two tasks: module observation and obstacle course navigation. Each task lasts for 1 minute, and the user has a timer in the left-hand corner, showing the time left to complete the task. In the module observation task, participants identify impacts marked by "M" or "W" stickers on panels identified by a letter and a number, e.g. A2 or B11. In the automatic control condition, the robot navigates autonomously, and the operator selects the correct impact. To trigger the automatic movement, the user needs to press one of the A1-A3 buttons, one per module. In the manual control

condition, the operator controls the robot's movement to inspect panels before making the selection.

The obstacle course task is designed to assess the operator's ability to navigate the robot through a series of obstacles between modules. In the automatic control condition, the robot moves forward autonomously, and the operator's task is to adjust the robot's orientation to avoid obstacles and impacts by rotating the robot as needed. In the manual control condition, the operator is responsible for both moving the robot forward through the obstacle course and controlling its orientation to avoid obstacles.

Data collected will include reports of identified impacts, task completion times, movement timestamps, and obstacle collisions. ECG readings will be recorded to monitor stress and engagement. A demographics and a perceived cognitive workload questionnaire will also be completed for background information.

2.1 Preliminary Laboratory Task

The first experimental campaign was conducted in the laboratory, where 15 participants (7 female, ages 18 to 32) completed the full task in a controlled environment. Each participant was asked to complete two automatic and two manual control circuits (training and trial), each consisting of navigating three modules and three obstacle courses.

In a future campaign, 15 divers will complete all three tasks—laboratory, underwater, and flight-based teleoperation—enabling intra-subject comparisons across different extreme environments.

2.2 Perception Flight Task

As of November 2024, a flight-based campaign is underway, focusing on both operator performance and system connectivity. In this phase, participants are seated in a plane and tasked with performing the perception module observation task. They view pre-recorded videos simulating the robot's movement through the modules, identifying impacts marked with the letters M or W on the panels. The flight experiment involves turns at 30° and 60° angles, with each participant experiencing 1.5g and 2g of gravitational force, respectively. Each video lasts 1 minute, during which participants must identify the correct impact letter.

This campaign is also used to measure the connectivity between the aircraft and the robot. This data will help establish a flight path for future tasks, where real-time communication with the robot on the ground will be required.

3 PRELIMINARY RESULTS

The preliminary analysis of the module observation task was done by measuring performance by counting the number of correct identifications of both the panel and the letter (e.g., identifying impacts labeled "M" or "W"). The maximum possible number of correct answers was 5. A one-way ANOVA comparing Control and Perception conditions showed no significant differences during the Training Phase ($M_{control}=3.24, M_{perception}=3.67, p=0.219$) or the Trial Phase ($M_{control}=4.09, M_{perception}=3.67, p=0.097$). However, when combining the two phases, a significant difference was found between the Training and Trial phases ($p=0.014$), indicating a learning effect. No significant differences were observed between Control and Perception conditions across phases ($p=0.074$), suggesting consistent operator performance irrespective of these conditions.

For the obstacle course, the ANOVA results show significant differences across all comparisons. Each obstacle course had a maximum time of 60s to be completed, even though extra time was allocated if the subject did not finish in time. The control condition had a higher mean total time ($M_{control}=59.24s$) compared to the perception condition ($M_{perception}=42.31s, p<0.001$), as the control task involved manually navigating five obstacles, while in perception, the robot moved autonomously, and participants only

needed to avoid two obstacles by altering its orientation. Similarly, training tasks had a higher mean time ($M_{training}=55.53s$) than trial tasks ($M_{trial}=46.01s, p<0.001$), consistent with a learning effect. Both control trial versus perception trial ($M_{control}=55.27s$ vs. $M_{perception}=36.75s, p<0.001$) and control training versus perception training ($M_{control}=63.21s$ vs. $M_{perception}=47.86s, p<0.001$) also showed significant differences, further emphasizing the impact of task design and operational demands on performance.

4 DISCUSSION

The preliminary results indicate that in the module task, the operators had similar accuracy between the Control and Perception conditions. However, a learning effect was observed as performance improved from the Training to the Trial Phase. In contrast, the obstacle course results showed that the Control condition (manual control with five obstacles) took longer to complete than the Perception condition (autonomous robot with two obstacles). This highlights the impact of task complexity and control mode on performance, with manual control requiring more time and effort.

Fully automated systems are highly effective in tasks requiring speed and precision, like disaster response or underwater exploration. However, operators can improve significantly with experience, highlighting the importance of robust training for adapting to manual control in complex scenarios. The results highlight the need for adaptable systems that dynamically allocate control based on task demands and operator stress. These systems could adjust interface complexity to prevent cognitive overload and prioritize automation in repetitive or high-stress tasks, while allowing operator intervention for nuanced decision-making in scenarios like teleoperation or remote medical procedures.

The integration of ECG data could enhance HMI design by providing real-time monitoring of heart rate and variability, key indicators of stress and cognitive load. Teleoperation interfaces could use this feedback to dynamically adapt, simplifying tasks during high stress or enabling finer manual control during calmer periods, thereby improving efficiency and reducing errors. Additionally, ECG analysis could inform tailored training programs by identifying stress points and cognitive overload, helping to build operator resilience and optimize performance in demanding environments.

This research explores the challenges of remote teleoperation in extreme conditions, combining controlled laboratory studies with real-world experiments. Initial results from the laboratory campaign highlight differences in task performance between automatic and manual control modes, providing a baseline. Future work could investigate how these baseline findings translate to predictive models that anticipate operator performance under stress, aiding in the development of systems that proactively adapt to operator needs in dynamic environments.

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